

Aditya Barman

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EDUCATION

University of Illinois Urbana-Champaign (UIUC)

Champaign, IL

Bachelor of Science in Computer Science & Statistics, Minor in Mathematics

Graduation: May 2027

- GPA: 4.0/4.0
- James Scholars Honors Program
- Activities: Pulse Competitions Committee, CS Sail Logistics Staff, Simplify Campus Ambassador

EXPERIENCE

Slack (Salesforce)

San Francisco, CA

Incoming Software Engineering Intern – Backend Engineering & AI Infrastructure

May 2026 – Aug. 2026

Motorola Solutions

Chicago, IL

Software Engineering Intern & Co-Op

May 2025 – Nov. 2025

- Delivered a new documentation access platform to 350 engineers by architecting a RAG system (Python, LangChain, ChromaDB, Gemini API) over 30K+ production documents.
- Enabled visual search across 10K+ engineering schematics by extending the RAG system with a multimodal CLIP vector embedding layer.
- Cut org-wide manual documentation effort 80% by shipping a Python DrawSVG diagram-generation service to production, automating engineering diagram creation across teams.

University of Illinois & United States Department of Agriculture (USDA)

Urbana, IL

Computer Vision & Machine Learning Researcher

Nov. 2024 – May 2025

- Extracted 1,700+ morphological measurements from 198 image pairs at 95%+ accuracy by engineering a Python computer vision pipeline (OpenCV, NumPy) for perspective warping, contour segmentation, and feature analysis.
- Accelerated USDA breeding program analysis by training cross-validated TensorFlow regression models to predict kernel expansion ratios and delivering automated Matplotlib reports to research stakeholders.

Chief Digital and Artificial Intelligence Office – Department of Defense

Washington, D.C.

Software Development Intern – AI4Defense Program

June 2024 – Aug. 2024

- Reduced manual policy review 90% across 500+ government documents by shipping a Python + OpenAI full-stack app with REST APIs for PDF ingestion, key-term extraction, and sentiment scoring.
- Earned a podium finish at the AI4Defense Showcase by building React + TypeScript dashboards with interactive visualizations that translated NLP outputs into narratives for senior policy analysts.

George Mason University (GMU)

Fairfax, VA

Artificial Intelligence Research Intern

June 2024 – Aug. 2024

- Quantified inter-rater reliability across 200+ algorithm grading tasks by benchmarking 4 LLM assistants (ChatGPT, Copilot, Claude, Gemini) using Intraclass Correlation Coefficients and statistical consistency metrics.
- Co-authored a research paper accepted to the MIT Undergraduate Research Technology Conference (URTC) 2024 on the feasibility of automated LLM-assistant-based academic assessment.

PROJECTS

Samur.ai | *Next.js, TypeScript, React, Tailwind CSS, Gemini API, Coolify*

Mar. 2026

- Cut average student schedule setup time from 30+ minutes to under 2 minutes by building an NLP scheduling engine (Next.js, TypeScript, Gemini API) that parses course syllabi and auto-generates optimized study plans.
- Self-deployed to production via Coolify on a personal VPS, owning CI/CD, environment configuration, and uptime monitoring end-to-end.

EchoBrief | *Python, FastAPI, React, OpenAI Whisper, Gemini API*

Sep. 2025

- Cut manual post-incident review 70% by architecting an event-driven audio pipeline (Python, FastAPI, OpenAI Whisper, Gemini API) that ingests real-time radio streams and extracts structured incident timelines.
- Placed Top 10 of 495 at Motorola's Open Innovation Hackathon by building a React + FastAPI interface letting dispatchers query transcribed audio for real-time summaries.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, SQL (PostgreSQL, MySQL), R

Frameworks: React, Next.js, Node.js, Express, Django, FastAPI, Flask, REST APIs, Prisma ORM, OAuth 2.0

Developer Tools: AWS (Certified Cloud Practitioner), GCP, Docker, Kubernetes, CI/CD, Git, Linux

Libraries: TensorFlow, LangChain, OpenCV, NumPy, Pandas, ChromaDB, Gemini API, OpenAI API